

The Formal Beginning: $(\mathbb{S}, \tau, \omega)$, the Unity Boundary, and the State as the Locus of Complexity

Internal working paper.

Succession. Predates the current $(\mathbb{S}, \tau)$ Foundation; superseded by it on the same terms as the other old-triple documents. χ here is the original “depth of deformation” parameter, retired along with the rest of the $(\mathbb{S}, \tau, \omega)$ apparatus — not to be confused with Becoming’s surviving $\chi_{\text{kin}}/\chi_{\text{sector}}/\chi_{\text{depth}}/\chi_{\text{shape}}$, a different, later taxonomy. THM 1–3 are absorbed into current Foundation §2. Two items remain live references despite the supersession: the term-for-term Gresnigt–Gourlay–Varma verification below (§4) and the CP-candidate mechanism (algebra-conjugation versus J_ω mismatch) — both cited forward, not reproduced, by later work.

1. The formal data

[DEF] The pattern. $\mathbb{S} = \mathbb{R}_\varepsilon[(\mathbb{Z}_2)^4]$, the sixteen-dimensional real Cayley–Dickson algebra: unital, flexible, quadratic, power-associative, central simple, non-alternative, non-composition, with positive-definite norm $N(x) = x\bar{x}$. $\text{Aut}(\mathbb{S}) = G_2 \times S_3$ (Brown 1967; Eakin–Sathaye 1990).

[DEF] The split. τ , an involution in the S_3 factor: $\mathbb{O} = \text{Fix}(\tau)$, $\mathbb{O}e_8 = \ker(\tau + 1)$. Three conjugate splits exist, related by the order-three automorphism μ .

[DEF] The state. ω , a faithful normal state on the operational algebra $\mathcal{M}$ built from the representable stratum of $(\mathbb{S}, \tau)$.

[AX P-Em] Emergence. The order of deduction is not the order of being. $\mathbb{S}$ is the persistence-form of a pre-algebraic ground outside all algebraic and metric vocabulary; ω is that ground’s sole imprint on the pattern; every existence and uniqueness claim below is scoped *among representable structures*.

$(\mathbb{S}, \tau, \omega)$ is the formal beginning. Nothing in what follows adds a fourth ingredient.

2. Emergent representation, unsharply

[PROP] Symmetric impurity. No admissible configuration is unsettled purely, and none is settled totally: writing χ for the state’s depth of deformation from the pattern’s symmetric reference, $0 < \chi < \chi_{\text{sup}}$ identically. Both bounds are ideal, unattained.

[PROP] The two horizons are of different kinds. The lower bound ($\chi = 0$) is excluded by **disjointness**: it is realized by a genuine, singular functional — outside the normal folium, at norm-distance exactly 2 from every physical state — approachable weak-*, never occupied. The upper bound (χ_{sup}) is excluded by **nonexistence**: total settlement would require a tracial state, and the properly infinite operational algebra admits none. The world cannot return to its origin because the origin lies in a different representation; it cannot complete because completion is not a state its own algebra can have.

[PROP] Scale is a settled-sector concept. Rest frequency is $\omega = m/\hbar$; absent condensed mass, no frequency scale exists, and only the conformal class of the emerging geometry is physical. Consequently: statements about rates, durations, or expansion in the unsettled regime are not false but unposed — the vocabulary they require is itself a product of settling, not available prior to it. Where such statements are made about that regime, they are retrospective: the settled clocks' backward reckoning of a dimensionless ratio, not a claim about how fast anything happened while unsettled.

Representation, in this sense, is emergent but never sharply so: the pattern is timeless throughout; what is graded is the state's own depth, approaching representability's fuller exercise without ever completing it and without ever having been fully absent.

3. Unity as boundary

The reference configuration named informally as “pure unity” is precisely the lower horizon of §2: an ideal point of the closure of the state space, $\text{Aut}(\mathbb{S})$ -symmetric, singular, disjoint from every physical state. Two corrections to looser prior framings, now load-bearing:

[COR] It is not a state the universe *was*. “Emergence” is constitutive, not temporal; crystallization denotes the atemporal fact of a settled imprint existing at all, not an event with a before. Unity horizon and settlement horizon are boundary markers of a scale, not termini of a history.

[COR] The natural instrument for measuring distance to this boundary — Araki relative entropy — does not apply to it directly. Araki's construction requires both arguments to be normal functionals; the horizon is singular by construction. Forcing the comparison through a limiting sequence of normal states generically diverges. The working measure of χ is therefore the primary (order-parameter) definition — aggregate condensate magnitude, ordinary finite expectation values — not relative entropy to the horizon. Norm-distance to the horizon is well-defined but constant at 2 for any physical state, and so is informative only as confirmation of disjointness, not as a varying measure of depth.

Unity, formally, is a limit no state occupies and no entropy formula reaches — present in the architecture as exactly enough structure to organize the scale, absent from it as anything more.

4. What real $\mathbb{S}$ cannot supply, proven

The question tested: does the triple, with $\mathbb{S}$ kept real, already contain everything the published algebraic construction of Standard-Model-shaped fermion content requires — or must the seed be complex from the outset?

[FACT, verified exactly] The corpus's ψ , its three generation octonions $\mathbb{O}_1, \mathbb{O}_2, \mathbb{O}_3$, their shared quaternionic core, and its full multiplication table are identical — not merely analogous — to the corresponding objects in Gresnigt, Gourlay & Varma (*Eur. Phys. J. C* 83, 2023) and its $C\ell(8)$ sequel (2024): the formulas match term for term, the table matches on all 256 entries. This is the literature the seed question is tested against.

[THM 1] No nonzero real element $A \in \mathbb{S}$ satisfies $A^2 = 0$. *Proof:* for pure-imaginary A , power-associativity gives $A^2 = -N(A)$; N is positive-definite.

[THM 2] No nonzero real $A \in \mathbb{S}$ satisfies $L_A^2 = 0$ as an operator, despite non-associativity making this a logically distinct condition from Theorem 1. *Proof:* $\text{tr}(L_A^2) = -16N(A)$ identically (established via the exact singular-value spectrum of L_v); writing $L_A^2 = -N(A)\text{Id} + T_A$ with $T_A(x) = (A, A, x)$ the associator, this forces $\text{tr}(T_A) = 0$ for every A — incompatible with the $T_A = N(A)\text{Id}$ that operator-nilpotency requires, unless $N(A) = 0$.

[FACT, verified] The realized Clifford structure of $\mathbb{S}$ ($C\ell(8)$, via left-multiplication by $e_1, \dots, e_8$; and the $\text{Im}(\mathbb{O})$ -intrinsic $C\ell(7)$, via $e_1, \dots, e_7$) are real algebras, consistent with real Bott periodicity. Their volume/chirality elements $\Gamma_8 = L_{e_1} \cdots L_{e_8}$ and $\gamma_7 = L_{e_1} \cdots L_{e_7}$ both square to $+1$: genuine chirality operators, splitting 16 real dimensions into two real 8-dimensional halves, but not complex structures. γ_7 is additionally central with respect to $e_1, \dots, e_7$ and anticommutes with every wild-side generator — a candidate formal home for the two-chiral-halves structure of the mass mechanism, independent of the present question.

[THM 3, general] Compatible orthogonal complex structures ($J^2 = -1$, orthogonal, skew) exist abundantly on any even-dimensional real inner-product space — pure linear algebra, no obstruction. What Theorems 1-2 forbid is specifically the realization of such a J as L_A for some $A \in \mathbb{S}$: even where an L_A happens to be one such J on a subspace, the global operator-nilpotency it would need to deliver still fails, by Theorem 2, everywhere.

Conclusion. $\mathbb{S}$'s own multiplication cannot double as the required Clifford/CAR product. This is a fact about the algebra, closed, general, and

independent of which generator or combination is tried.

5. Where complexity is licensed to live

Theorem 3 identifies the actual content of the obstruction: not that complex structure is unavailable, but that it cannot be *algebra-intrinsic*. This is precisely the distinction the jurisdiction axiom already draws.

[AX, jurisdiction] Exact, universal features belong to the pattern; generic, state-dependent features belong to the imprint. A complex structure adjoined to $\mathbb{S}$ directly would be exact and universal — present in the definition of the algebra, the same for every configuration, asserted rather than derived — exactly what Theorems 1-2 prove $\mathbb{S}$'s own arithmetic cannot supply on its own terms. A complex structure supplied by ω is generic and contingent — selected by a physically motivated dynamics, potentially different for different configurations — and carries none of the same requirement, because nothing requires it to equal any L_A .

[PROP, proven mechanism] This is not merely permitted; it is already demonstrated working, once, for the bosonic sector. Given a real classical system and a chosen flow, the standard one-particle-structure construction (Kay-Wald) produces a genuine complex Hilbert space and a KMS state with no complex structure adjoined by hand: real phase space, chosen dynamics, complex structure derived. The corpus's own squeeze-thermal-state construction is exactly this, executed and verified: the squeeze flow on the octonionic scale line yields $H_1 = L^2(\mathbb{R}_u) \otimes L^2(S^2)$, a unique regular β -KMS state, and a proven type III₁ verdict — a real system's own flow producing complexity, not receiving it.

[COR] Every occurrence of “*i*” in this program is now located precisely: not in the pattern, never in $\mathbb{S}$'s own multiplication, but in the state's flow-selected positive-frequency structure, wherever that structure is required. The seed remains $(\mathbb{S}, \tau, \omega)$, real, unmodified.

6. The natural extension: fermions on the same base

[PROP, proven, prior result] The octonionic product on $\text{Im}(\mathbb{O})$ splits, symmetric and antisymmetric halves, into two structures from one multiplication: the antisymmetric (Malcev) half is the R-flux bracket generating X, P, W — the bosonic sector's own foundation; the symmetric (Jordan) half is $a \circ a = -|a|^2 e_0$, an ordinary positive-definite quadratic form, verified exactly. One space, one product, two channels.

This identifies the economical candidate base for a fermionic construction: not a freshly invented space, but $\text{Im}(\mathbb{O})$'s own Jordan channel — the same seven-

dimensional space already housing X, P, W , its quadratic structure already present and proven, requiring no new axiom.

[FACT, verified] Testing the natural chirality element of this space’s own realized Clifford structure (γ_7 , §4) confirms it is real, not complex — consistent with, and independently confirming, Theorem 3: no algebra-intrinsic shortcut is available here either. What remains is the standard route: an abstract real Clifford algebra over this quadratic form, a state-selected flow, and the resulting representation-level complex structure.

[TARGET, named, three-part] 1. The correct scaling weight for a Clifford-valued (rather than scalar) field under the already-proven squeeze/Mellin construction — the change-of-variables argument making squeeze equal translation along the scale line is a fact about the flow, not about what is being scaled, and is expected to extend. 2. The fermionic positivity and uniqueness analysis, using Rocca-Sirugue-Testard’s fermionic classification (*Commun. Math. Phys.* 13, 1969) — already cited in the corpus’s own bibliography, unused until now. Its stated result parallels the bosonic case exactly: a one-particle evolution generally forces a unique quasi-free equilibrium state, with non-uniqueness excluded by a named criterion (absence of a Fock part) directly analogous to the zero-mode door already shut for bosons. 3. Comparison of the resulting nilpotent creation operators, restricted to one generation’s real subspace, against the verified Gresnigt formulas of §4.

None of this is complete. All of it is now a bounded task with named tools, not an open question of possibility.

7. Texture, and the site of asymmetry

[FACT] No finite subgroup of $\text{Aut}(\mathbb{S})$ can act transitively on the infinite-dimensional space of normal states. The operative symmetry there is $\text{Ad}(U(\mathcal{M}))$, and by Connes-Størmer homogeneity it achieves only approximate transitivity: any two normal states can be brought arbitrarily close, never made identical.

[COR] This is the formal ground on which irreducible, non-symmetry-reducible content is permitted to be real at all — including, if it exists, a fixed asymmetry between the algebra’s own conjugation and the state’s modular conjugation J_ω . That specific mismatch remains the standing candidate mechanism for CP violation; Connes-Størmer explains why room for such a mismatch exists in the representation, not what its value is. **Update:** the bosonic-sector precursor to this question — J_ω ’s one-particle shadow against the canonical reflections $(X, -P)$, $(-X, P)$, the well-posed comparison Permission Slip narrows this to at that level — is now answered: alignment, exact at every temperature, no mismatch found (J_ω Resolved, companion note; two

independent constructions, pending external check). This closes the bosonic precursor cleanly but does not itself answer the question posed here, which concerns the algebra's own conjugation against J_ω in a sense requiring the fermionic sector to be well-posed in full — the bosonic door is shown clean; the fermionic door, where any genuine mismatch would have to live, remains open and is now the sole remaining site for this mechanism.

Prepared as an internal record of $(\mathbb{S}, \tau, \omega)$'s current status: proven where marked, conjectured where marked, with the failed algebra-intrinsic routes closed and not revisited.